

Notes on Uniform Law of Large Numbers

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1 Empirical Process Theory

1.1 Abstract Glivenko-Cantelli Theorem (Theorem 19.4 of Van der Vaart (1998) or Theorem 2.4.1 of Van der Vaart & Wellner (1996))

Every class $\mathcal{F}$ of measurable functions such that $N_{[]}(\varepsilon, \mathcal{F}, L_1(P)) < \infty$ for every $\varepsilon > 0$ is P-Glivenko-Cantelli i.e., $\|\mathbb{P}_n - P\|_{\mathcal{F}} := \sup_{f \in \mathcal{F}} |\mathbb{P}_n f - P f| \rightarrow_{as} 0$.

Proof:

Given a $\varepsilon > 0$, as $N_{[]}(\varepsilon, \mathcal{F}, L_1(P)) < \infty$ by hypothesis i.e., the minimum number of ε -brackets with $L_1(P)$ norm needed to cover $\mathcal{F}$ is finite, there is a finite sequence of ε -brackets with $L_1(P)$ norm such that $\mathcal{F} \subseteq \cup_i [l_i, u_i]$ where $\|u_i - l_i\|_{P,1} := (P|u_i - l_i|^1)^{1/1} = P(u_i - l_i) < \varepsilon$. For any $f \in \mathcal{F}$ and any i in the finite sequence,

$$(\mathbb{P}_n - P)f = \mathbb{P}_n f - P u_i + P u_i - P f \leq (\mathbb{P}_n - P)u_i + P(u_i - f) \leq (\mathbb{P}_n - P)u_i + P(u_i - l_i) < (\mathbb{P}_n - P)u_i + \varepsilon$$

since $l_i \leq f \leq u_i$. Thus,

$$\sup_{f \in \mathcal{F}} (\mathbb{P}_n - P)f < \max_j (\mathbb{P}_n - P)u_j + \varepsilon$$

Given x , $\max_j (\mathbb{P}_n - P)u_j(x) \rightarrow_{as} 0$ as $n \rightarrow \infty$ by strong law of large numbers for a sequence of iid random variables. Similarly,

$$(\mathbb{P}_n - P)f = \mathbb{P}_n f - P l_i + P l_i - P f \geq (\mathbb{P}_n - P)l_i + P(l_i - f) \geq (\mathbb{P}_n - P)l_i + P(l_i - u_i) > (\mathbb{P}_n - P)u_i - \varepsilon$$

Thus,

$$\inf_{f \in \mathcal{F}} (\mathbb{P}_n - P)f > \min_j (\mathbb{P}_n - P)u_j - \varepsilon$$

Given x , $\min_j (\mathbb{P}_n - P)u_j(x) \rightarrow_{as} 0$ as $n \rightarrow \infty$ by strong law of large numbers for a sequence of iid random variables. Together,

$$-\varepsilon < \lim_{n \rightarrow \infty} \inf_{f \in \mathcal{F}} (\mathbb{P}_n - P)f \leq \lim_{n \rightarrow \infty} \sup_{f \in \mathcal{F}} (\mathbb{P}_n - P)f < \varepsilon \quad a.s.$$

Therefore, $|\lim_{n \rightarrow \infty} \sup_{f \in \mathcal{F}} (\mathbb{P}_n - P)f| = \lim_{n \rightarrow \infty} \sup_{f \in \mathcal{F}} |(\mathbb{P}_n - P)f| := \lim_{n \rightarrow \infty} \|\mathbb{P}_n - P\|_{\mathcal{F}} < \varepsilon$ a.s.

Q.E.D.

1.2 Corollary 1 (Pointwise Compact Class) (Example 19.8 of Van der Vaart (1998))

If $\mathcal{F} = \{f_\theta : \theta \in \Theta\}$ where (i) f_θ is a measurable and continuous function of θ . (ii) The envelope (i.e., dominating) function of f is integrable. (iii) Θ is a compact set. Then, $N_{[]}(\varepsilon, \mathcal{F}, L_1(P)) < \infty$ (and $\sup_{f \in \mathcal{F}} |\mathbb{P}_n f - P f| \rightarrow_{as} 0$ by Abstract Glivenko-Cantelli Theorem)

Proof:

Let $B_i \subset \Theta$ with $\text{center}(B_i)$ $\theta_i \in \Theta$ and B_i is an open set. We construct the bracket $[f_{B_i}, f^{B_i}]$ where $f_{B_i} := \inf_{\theta \in B_i} f(\theta)$ and $f^{B_i} := \sup_{\theta \in B_i} f(\theta)$,

$$\begin{aligned} \|f_{B_i}, f^{B_i}\|_{P,1} &= (P|f^{B_i} - f_{B_i}|^1)^{1/1} \\ &= P(f^{B_i} - f_{B_i}) \\ &= Pf^{B_i} - Pf_{B_i} \\ &= \int f^{B_i} dP - \int f_{B_i} dP \end{aligned}$$

Let $\{B_{im}\}_{m=1}^\infty$ be the sequence of B_i with the same center θ_i and the radius of them converges to zero as $m \rightarrow \infty$. Thus,

$$\begin{aligned} \lim_{m \rightarrow \infty} \int f^{B_{im}} dP - \lim_{m \rightarrow \infty} \int f_{B_{im}} dP &= \int \lim_{m \rightarrow \infty} f^{B_{im}} dP - \int \lim_{m \rightarrow \infty} f_{B_{im}} dP \\ &\quad \text{by Dominated Convergence Theorem with (ii)} \\ &= \int f^{\lim_{m \rightarrow \infty} B_{im}} dP - \int f_{\lim_{m \rightarrow \infty} B_{im}} dP \quad \text{by (i)} \\ &= \int f_{\theta_i} dP - \int f_{\theta_i} dP = 0 \end{aligned}$$

Therefore, the bracket $[f_{B_i}, f^{B_i}]$ has $L_1(P)$ -size ε for any $\varepsilon > 0$. As θ_i is arbitrary from Θ , we have $\{[f_{B_i}, f^{B_i}]\}_{i=1}^\infty$. Its corresponding $\{B_i\}_{i=1}^\infty$ is a cover of Θ because $\cup_{i=1}^\infty B_i = \Theta$ where $B_i \subset \Theta$. By hypothesis (iii), $\{B_i\}_{i=1}^\infty$ has a finite subcover that also covers Θ i.e., we have $\{B_j\}_{j=1}^p \subset \{B_i\}_{i=1}^\infty$ such that $\cup_{j=1}^p B_j = \Theta$ where $p < \infty$. This implies that the minimum number of ε -brackets with $L_1(P)$ norm needed to cover $\mathcal{F}$ is finite i.e., $N_{[]}(\varepsilon, \mathcal{F}, L_1(P)) < \infty$ Q.E.D.

Remark 1: The consistency of M-estimator e.g., Theorem 5.7 of Van der Vaart (1998) usually (not a must) requires the uniform convergence $\sup_{f \in \mathcal{F}} |\mathbb{P}_n f - Pf| \rightarrow_{as} \sup_{\theta \in \Theta} |\mathbb{P}_n m_\theta - P m_\theta| \rightarrow_{as} 0$. This example provides the sufficient conditions.

Remark 2: This form of uniform law of large numbers can be derived directly without using the Abstract Glivenko-Cantelli Theorem.

1.3 Uniform Law of Large Numbers (Direct Derivation, Theorem 4.2.1 of Amemiya (1985))

Let $g(y; \theta)$ be a measurable function of y for each $\theta \in \Theta$, which is a compact subset of $\mathbb{R}^K$, and a continuous function of $\theta \in \Theta$ for each y . Assume $\mathbb{E}[g(y; \theta)] = 0$. Let $\{y_i\}_{i=1}^\infty$ be a sequence of i.i.d. random variables such that $\mathbb{E}[\sup_{\theta \in \Theta} |g(y; \theta)|] < \infty$ (the dominance condition). Then, $\sup_{\theta \in \Theta} |N^{-1} \sum_{i=1}^N g(y_i; \theta)| \rightarrow_p 0$ as $N \rightarrow \infty$.

Proof:

We have $\{\Theta_j^M\}_{j=1}^M$ where $\cup_{j=1}^M \Theta_j^M = \Theta$ and $\Theta_j^M \cap \Theta_k^M = \emptyset$ for any $j \neq k$ and $d(\Theta_j^M, \Theta_k^M) \rightarrow 0$ as $M \rightarrow \infty$. Let $\theta_j \in \Theta_j^M$, $j = 1, \dots, M$. Define $g_i(\theta) = g(y_i; \theta)$ for convenience. For any $\varepsilon > 0$,

$$\sup_{\theta \in \Theta} |N^{-1} \sum_{i=1}^N g_i(\theta)| > \varepsilon \implies \cup_{j=1}^M \{\sup_{\theta \in \Theta_j^M} |N^{-1} \sum_{i=1}^N g_i(\theta)| > \varepsilon\}$$

$$\begin{aligned}
Pr(\sup_{\boldsymbol{\theta} \in \Theta} |N^{-1} \sum_{i=1}^N g_i(\boldsymbol{\theta})| > \varepsilon) &\leq Pr(\cup_{j=1}^M \{ \sup_{\boldsymbol{\theta} \in \Theta_j^M} |N^{-1} \sum_{i=1}^N g_i(\boldsymbol{\theta})| > \varepsilon \}) \\
&\leq \sum_{j=1}^M Pr(\sup_{\boldsymbol{\theta} \in \Theta_j^M} |N^{-1} \sum_{i=1}^N g_i(\boldsymbol{\theta})| > \varepsilon) \quad \text{as } Pr(A \cup B) = Pr(A) + Pr(B) - Pr(A \cap B) \\
&= \sum_{j=1}^M Pr(\sup_{\boldsymbol{\theta} \in \Theta_j^M} |N^{-1} \sum_{i=1}^N g_i(\boldsymbol{\theta}) + N^{-1} \sum_{i=1}^N g_i(\boldsymbol{\theta}_j) - N^{-1} \sum_{i=1}^N g_i(\boldsymbol{\theta}_j)| > \varepsilon) \\
&= \sum_{j=1}^M Pr(\sup_{\boldsymbol{\theta} \in \Theta_j^M} |N^{-1} \sum_{i=1}^N g_i(\boldsymbol{\theta}_j) + N^{-1} \sum_{i=1}^N (g_i(\boldsymbol{\theta}) - g_i(\boldsymbol{\theta}_j))| > \varepsilon)
\end{aligned}$$

$$\text{As } |a+b| > \varepsilon \implies \{|a| > \varepsilon/2\} \cup \{|b| > \varepsilon/2\}, Pr(|a+b| > \varepsilon) \leq Pr(\{|a| > \varepsilon/2\} \cup \{|b| > \varepsilon/2\}) \leq Pr(|a| > \varepsilon/2) + Pr(|b| > \varepsilon/2)$$

$$\begin{aligned}
&\leq \sum_{j=1}^M \{Pr(\sup_{\boldsymbol{\theta} \in \Theta_j^M} |N^{-1} \sum_{i=1}^N g_i(\boldsymbol{\theta}_j)| > \varepsilon/2) + Pr(\sup_{\boldsymbol{\theta} \in \Theta_j^M} |N^{-1} \sum_{i=1}^N (g_i(\boldsymbol{\theta}) - g_i(\boldsymbol{\theta}_j))| > \varepsilon/2)\} \\
&= \sum_{j=1}^M Pr(|N^{-1} \sum_{i=1}^N g_i(\boldsymbol{\theta}_j)| > \varepsilon/2) + \sum_{j=1}^M Pr(\sup_{\boldsymbol{\theta} \in \Theta_j^M} |N^{-1} \sum_{i=1}^N (g_i(\boldsymbol{\theta}) - g_i(\boldsymbol{\theta}_j))| > \varepsilon/2) \\
&= \sum_{j=1}^M Pr(|N^{-1} \sum_{i=1}^N g_i(\boldsymbol{\theta}_j)| > \varepsilon/2) + \sum_{j=1}^M Pr(N^{-1} \sum_{i=1}^N \sup_{\boldsymbol{\theta} \in \Theta_j^M} |g_i(\boldsymbol{\theta}) - g_i(\boldsymbol{\theta}_j)| > \varepsilon/2)
\end{aligned}$$

by triangle inequality

Now consider applying law of large numbers to $N^{-1} \sum_{i=1}^N \sup_{\boldsymbol{\theta} \in \Theta_j^M} |g_i(\boldsymbol{\theta}) - g_i(\boldsymbol{\theta}_j)| \rightarrow_p \mathbb{E}[\sup_{\boldsymbol{\theta} \in \Theta_j^M} |g_i(\boldsymbol{\theta}) - g_i(\boldsymbol{\theta}_j)|]$ as $N \rightarrow \infty$,

$$\begin{aligned}
\sup_{\boldsymbol{\theta} \in \Theta_j^M} |g_i(\boldsymbol{\theta}) - g_i(\boldsymbol{\theta}_j)| &\leq \sup_{\boldsymbol{\theta} \in \Theta_j^M} |g_i(\boldsymbol{\theta})| + \sup_{\boldsymbol{\theta} \in \Theta_j^M} |g_i(\boldsymbol{\theta}_j)| \quad \text{by triangle inequality} \\
&= \sup_{\boldsymbol{\theta} \in \Theta_j^M} |g_i(\boldsymbol{\theta})| + |g_i(\boldsymbol{\theta}_j)| \\
&\leq \sup_{\boldsymbol{\theta} \in \Theta_j^M} |g_i(\boldsymbol{\theta})| + \sup_{\boldsymbol{\theta} \in \Theta_j^M} |g_i(\boldsymbol{\theta})| \quad \text{as } \boldsymbol{\theta}_j \in \Theta_j^M \\
&= 2 \sup_{\boldsymbol{\theta} \in \Theta_j^M} |g_i(\boldsymbol{\theta})| \\
&\leq 2 \sup_{\boldsymbol{\theta} \in \Theta} |g_i(\boldsymbol{\theta})|
\end{aligned}$$

The dominance condition hypothesized implies $\sup_{\boldsymbol{\theta} \in \Theta} |g_i(\boldsymbol{\theta})|$ is integrable. Thus,

$$\begin{aligned}
\lim_{M \rightarrow \infty} \mathbb{E}[\sup_{\boldsymbol{\theta} \in \Theta_j^M} |g_i(\boldsymbol{\theta}) - g_i(\boldsymbol{\theta}_j)|] &= \mathbb{E}[\lim_{M \rightarrow \infty} \sup_{\boldsymbol{\theta} \in \Theta_j^M} |g_i(\boldsymbol{\theta}) - g_i(\boldsymbol{\theta}_j)|] \quad \text{by Dominated Convergence Theorem} \\
&= \mathbb{E}[0] \quad \text{by continuity of } g \\
&= 0
\end{aligned}$$

uniformly for j . Picking a large enough M such that $\mathbb{E}[\sup_{\boldsymbol{\theta} \in \Theta_j^M} |g_i(\boldsymbol{\theta}) - g_i(\boldsymbol{\theta}_j)|] \leq \varepsilon/2$, $Pr(N^{-1} \sum_{i=1}^N \sup_{\boldsymbol{\theta} \in \Theta_j^M} |g_i(\boldsymbol{\theta}) - g_i(\boldsymbol{\theta}_j)| > \varepsilon/2) \rightarrow 0$ as $N \rightarrow \infty$.

Now consider applying law of large numbers to $N^{-1} \sum_{i=1}^N g_i(\boldsymbol{\theta}_j) \rightarrow_p \mathbb{E}[g_i(\boldsymbol{\theta}_j)] = 0$ as $N \rightarrow \infty$. This implies $Pr(|N^{-1} \sum_{i=1}^N g_i(\boldsymbol{\theta}_j)| > \varepsilon/2) \rightarrow 0$ as $N \rightarrow \infty$. Together, $Pr(\sup_{\boldsymbol{\theta} \in \Theta} |N^{-1} \sum_{i=1}^N g_i(\boldsymbol{\theta})| > \varepsilon) = Pr(|\sup_{\boldsymbol{\theta} \in \Theta} |N^{-1} \sum_{i=1}^N g_i(\boldsymbol{\theta})| - 0| > \varepsilon) \rightarrow 0$ as $N \rightarrow \infty$.
i.e., $\sup_{\boldsymbol{\theta} \in \Theta} |N^{-1} \sum_{i=1}^N g_i(\boldsymbol{\theta})| \rightarrow_p 0$ as $N \rightarrow \infty$. Q.E.D.

Remark 1: The assumption $\mathbb{E}[g(y; \boldsymbol{\theta})] = 0$ is made without loss of generality. It is because we can replace $g(y; \boldsymbol{\theta})$ with $a(y; \boldsymbol{\theta}) - \mathbb{E}[a(y; \boldsymbol{\theta})]$ given the fact that if $a(y; \boldsymbol{\theta})$ is continuous in $\boldsymbol{\theta}$ and satisfies the dominance condition i.e., bounded by integrable dominating function e.g., $\mathbb{E}[\sup_{\boldsymbol{\theta} \in \Theta} |a(y; \boldsymbol{\theta})|] < \infty$, then $\mathbb{E}[a(y; \boldsymbol{\theta})]$ is continuous in $\boldsymbol{\theta}$.

$$\begin{aligned}
\lim_{\boldsymbol{\theta} \rightarrow \boldsymbol{\theta}'} \mathbb{E}[a(y; \boldsymbol{\theta})] &= \mathbb{E}[\lim_{\boldsymbol{\theta} \rightarrow \boldsymbol{\theta}'} a(y; \boldsymbol{\theta})] \quad \text{by Dominated Convergence Theorem} \\
&= \mathbb{E}[a(y; \boldsymbol{\theta}')] \quad \text{by continuity of } a
\end{aligned}$$

Remark 2: Lemma 2.4 of Newey & McFadden (1994) weakens the continuity assumption to continuity with probability one. The proof is neat and can be found in the appendix of Tauchen (1985).

Remark 3: The convergence in probability here can be strengthened to almost sure convergence with the same hypothesis (see Appendix).

1.4 Corollary 2 (Distribution function) (Example 19.6 of Van der Vaart (1998))

If $\mathcal{F} = \{\mathbb{1}(x \in (-\infty, t]), t \in \mathbb{R}\}$, Abstract Glivenko-Cantelli Theorem reduces to the (classical) Glivenko-Cantelli Theorem i.e., $\|\mathbb{F}_n - F\|_\infty := \sup_{t \in \mathbb{R}} |\mathbb{F}_n(t) - F(t)| \rightarrow_{as} 0$.

Proof:

Given a $\varepsilon > 0$, there is a grid of points $-\infty = t_0 < t_1 < \dots < t_k = \infty$ such that $F(t_i-) - F(t_{i-1}) < \varepsilon$ for $\forall i$ (only points at the partition can jump more than ε). We construct the bracket $[\mathbb{1}(x \in (-\infty, t_{i-1}]), \mathbb{1}(x \in (-\infty, t_i))]$,

$$\begin{aligned} \|\mathbb{1}(x \in (-\infty, t_i)) - \mathbb{1}(x \in (-\infty, t_{i-1}))\|_{F,1} &:= F(\mathbb{1}(x \in (-\infty, t_i)) - \mathbb{1}(x \in (-\infty, t_{i-1}))) \\ &= F(t_i-) - F(t_{i-1}) < \varepsilon \end{aligned}$$

Therefore, the bracket $[\mathbb{1}(x \in (-\infty, t_{i-1}]), \mathbb{1}(x \in (-\infty, t_i))]$ has $L_1(F)$ -size ε .

Q.E.D.

1.5 Classical Glivenko-Cantelli Theorem (Direct Derivation, Theorem 19.1 of Van der Vaart (1998))

If $\{X_i\}_{i=1}^\infty$ is i.i.d with distribution function F . Then $\|\mathbb{F}_n - F\|_\infty := \sup_{t \in \mathbb{R}} |\mathbb{F}_n(t) - F(t)| \rightarrow_{as} 0$.

Proof:

Given a $\varepsilon > 0$, there is a grid of points $-\infty = t_0 < t_1 < \dots < t_k = \infty$ such that $F(t_i-) - F(t_{i-1}) < \varepsilon$ for $\forall i$ (only points at the partition can jump more than ε). For $t_{i-1} \leq t < t_i$, note that $\mathbb{F}_n(t) \leq \mathbb{F}_n(t_i-)$ and $F(t_i-) - F(t_{i-1}) < \varepsilon \iff F(t_i-) - F(t) < \varepsilon \iff -F(t) < \varepsilon - F(t_i-)$

$$\mathbb{F}_n(t) - F(t) \leq \mathbb{F}_n(t_i-) - F(t) < \mathbb{F}_n(t_i-) - F(t_i-) + \varepsilon$$

Note also that $\mathbb{F}_n(t) \geq \mathbb{F}_n(t_{i-1})$ and $F(t_i-) - F(t_{i-1}) < \varepsilon \iff F(t) - F(t_{i-1}) < \varepsilon \iff F(t) < F(t_{i-1}) + \varepsilon$

$$\mathbb{F}_n(t) - F(t) \geq \mathbb{F}_n(t_{i-1}) - F(t) > \mathbb{F}_n(t_{i-1}) - F(t_{i-1}) - \varepsilon$$

For each $i \in \{1, \dots, k-1\}$, $\{\mathbb{F}_n(t_i-) - F(t_i-)\}_{n=1}^\infty$ and $\{\mathbb{F}_n(t_{i-1}) - F(t_{i-1})\}_{n=1}^\infty$ are sequences of i.i.d. random variables, both converge to zero almost surely by strong law of large numbers. Therefore,

$$|\lim_{n \rightarrow \infty} \sup_{t \in \mathbb{R}} (\mathbb{F}_n(t) - F(t))| = \lim_{n \rightarrow \infty} \sup_{t \in \mathbb{R}} |\mathbb{F}_n(t) - F(t)| := \lim_{n \rightarrow \infty} \|\mathbb{F}_n - F\|_\infty < \varepsilon \quad a.s.$$

Q.E.D.

1.6 Corollary 3 (Parametric Class, Example 19.7 of Van der Vaart (1998))

If $\mathcal{F} = \{f_\theta : \theta \in \Theta\}$ where (i) f_θ is a measurable function satisfying Lipschitz continuity i.e., there exists a measurable function m such that $|f_{\theta_1}(x) - f_{\theta_2}(x)| \leq m(x) \|\theta_1 - \theta_2\|$ for every θ_1, θ_2 . (ii) Θ is a bounded set and is a subset of $\mathbb{R}^d$. (iii)

$P|m|^r < \infty$. Then, there exists a $K(\Theta, d)$ such that $N_{[]} (2\varepsilon \|m\|_{P,r}, \mathcal{F}, L_r(P)) \leq K(\text{diam}(\Theta)/(2\varepsilon))^d$ for $0 < 2\varepsilon < \text{diam}(\Theta)$ (and $\sup_{f \in \mathcal{F}} |\mathbb{P}_n f - Pf| \rightarrow_{as} 0$ by Abstract Glivenko-Cantelli Theorem)

Proof:

Given a $0 < \varepsilon < \text{diam}(\Theta)/2$, we pick an arbitrary θ_1 from Θ , for any θ_2 such that $\|\theta_1 - \theta_2\| \leq \varepsilon$ i.e., from the closed ball with center θ_1 and radius ε , we have

$$|f_{\theta_1}(x) - f_{\theta_2}(x)| \leq m(x) \|\theta_1 - \theta_2\| \leq m(x) \varepsilon$$

Thus,

$$\begin{aligned} -m(x)\varepsilon &\leq f_{\theta_1}(x) - f_{\theta_2}(x) \leq m(x)\varepsilon \\ -f_{\theta_1}(x) - m(x)\varepsilon &\leq -f_{\theta_2}(x) \leq -f_{\theta_1}(x) + m(x)\varepsilon \\ f_{\theta_1}(x) + m(x)\varepsilon &\geq f_{\theta_2}(x) \geq f_{\theta_1}(x) - m(x)\varepsilon \\ f_{\theta_1}(x) - m(x)\varepsilon &\leq f_{\theta_2}(x) \leq f_{\theta_1}(x) + m(x)\varepsilon \end{aligned}$$

Therefore, we construct the bracket $[f_\theta - \varepsilon m, f_\theta + \varepsilon m]$ with $L_r(P)$ -size

$$\begin{aligned} \|f_\theta + \varepsilon m - (f_\theta - \varepsilon m)\|_{P,r} &= \|2\varepsilon m\|_{P,r} \\ &= (P|2\varepsilon m|^r)^{1/r} \\ &= (|2\varepsilon|^r P|m|^r)^{1/r} \\ &= 2\varepsilon \|m\|_{P,r} \end{aligned}$$

As the θ is arbitrary, we have $\{[f_\theta - \varepsilon m, f_\theta + \varepsilon m]\}_{\theta \in \Theta}$.

The largest size of Θ is $\text{diam}(\Theta) < \infty$ for any $d > 0$ (it is finite as Θ is bounded), the diameter of the closed ball defined above is 2ε . The largest number of closed balls needed to “cover” Θ is $(\text{diam}(\Theta)/(2\varepsilon))^d$ for any $d > 0$. Q.E.D.

Remark 1: Instead of assuming compact Θ in Corollary 1, Θ here is only bounded. However, the continuity assumption in Corollary 1 is strengthened to Lipschitz continuity. This example shows that the conditions of the typical uniform law of large numbers can be altered by using the Abstract Glivenko-Cantelli Theorem.

1.7 Uniform Law of Large Numbers with Bounded Parameter Space (Direct Derivation, Theorem 7.9 and 7.11 of Hansen (2020))

If $\{X_i\}_{i=1}^\infty$ is i.i.d. (i) $\mathbb{E}[|g(X_1; \theta)|] < \infty$ for $\forall \theta \in \Theta$ (ii) Θ is a bounded set (iii) $g(x; \theta)$ is first-order Lipschitz. Then, $\sup_{\theta \in \Theta} |N^{-1} \sum_{i=1}^N g(X_i; \theta) - \mathbb{E}[g(X_1; \theta)]| \rightarrow_p 0$.

Proof:

Combining Theorem 7.9 and 7.11 of Hansen (2020). The proof of Theorem 7.9 follows similar ideas of that of Theorem 4.2.1 of Amemiya (1985). Q.E.D.

Remark: (i) and (iii) of Corollary 3 implies that the function f_θ is first-order Lipschitz with $r = 1$. Corollary 3 does not require $\mathbb{E}[|g(X_1; \theta)|] < \infty$ (needed for the application of weak law of large numbers) and the convergence is stronger.

2 Reference

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3 Appendix

3.1 Proof of Strong Law of Large Numbers

3.1.1 Lemma 1

$$\sum_{i=1}^{\infty} Pr(|X| \geq i) \leq \mathbb{E}[|X|] \leq 1 + \sum_{i=1}^{\infty} Pr(|X| \geq i)$$

Proof: see Theorem 3.2.1 of Chung (2001).

Q.E.D.

3.1.2 Borel-Cantelli Theorem

Let $\{A_i\}_{i=1}^{\infty}$ be a sequence of events on the probability space,

- (a) If $\sum_{i=1}^{\infty} Pr(A_i) < \infty$, then $Pr(A_i \text{ i.o.}) = 0$;
- (b) If A_i is independent and $\sum_{i=1}^{\infty} Pr(A_i) = \infty$, then $Pr(A_i \text{ i.o.}) = 1$.

Proof: see Rao (1973, p.137-138).

Q.E.D.

3.1.3 Kolmogorov Theorem 1

Let $\{X_i\}_{i=1}^{\infty}$ be a sequence of independent (not necessarily identically distributed) random variables such that $\mu_i := \mathbb{E}[X_i]$ and $\sigma_i^2 := Var(X_i)$, we have

$$\sum_{i=1}^{\infty} \frac{\sigma_i^2}{i^2} < \infty \implies N^{-1} \sum_{i=1}^N (X_i - \mu_i) \rightarrow_{as} 0$$

Proof: see Rao (1973, p.114). It is a consequence of Hajek-Renyi inequality.

Q.E.D.

3.1.4 Kolmogorov Theorem 2

This theorem is frequently used in the proofs of this note so we prove it here for the sake of completeness.

Let $\{X_i\}_{i=1}^{\infty}$ be a sequence of i.i.d. random variables, we have

$$\mu := \mathbb{E}[X_1] < \infty \iff N^{-1} \sum_{i=1}^N X_i \rightarrow_{as} \mu$$

Proof:

($\Leftarrow$) (This part of the proof is from Rao (1973, p.115))

$$\begin{aligned} \frac{X_N}{N} &= N^{-1} \sum_{i=1}^N X_i - \frac{N-1}{N} (N-1)^{-1} \sum_{i=1}^{N-1} X_i \\ &\rightarrow_{as} \mu - 1 \cdot \mu = 0 \end{aligned} \quad \text{as } N^{-1} \sum_{i=1}^N X_i \rightarrow_{as} \mu$$

This implies that $Pr(|\frac{X_N}{N}| \geq 1 \text{ i.o.}) = Pr(|X_N| \geq N \text{ i.o.}) = 0$ where *i.o.* means “infinitely often”. Moreover, the independence of X_i implies the independence of the event $|X_N| \geq N$. Therefore, by Borel-Cantelli Theorem (using the logic If not B then not A for (b)), we have

$$\sum_{N=1}^{\infty} Pr(|X_N| \geq N) < \infty$$

By Lemma 1 above, we have $\mathbb{E}[|X_N|] < \infty$. This implies that $\mu := \mathbb{E}[X_1] < \infty$.

($\implies$) (This part of the proof is from Theorem 5.4.2 of Chung (2001))

Define

$$Y_i := \begin{cases} X_i & \text{if } |X_i| \leq i \\ 0 & \text{if } |X_i| > i \end{cases}$$

We apply $\{Y_i - \mathbb{E}[Y_i]\}_{i=1}^\infty$ to Kolmogorov Theorem 1 (note that it is not identically distributed and $\text{Var}(Y_i - \mathbb{E}[Y_i]) = \text{Var}(Y_i)$).

$$\begin{aligned} \sum_{i=1}^{\infty} \frac{\text{Var}(Y_i)}{i^2} &= \sum_{i=1}^{\infty} \frac{\mathbb{E}[Y_i^2] - \mathbb{E}[Y_i]^2}{i^2} \\ &\leq \sum_{i=1}^{\infty} \frac{\mathbb{E}[Y_i^2]}{i^2} \\ &= \sum_{i=1}^{\infty} \frac{1}{i^2} \int_{-\infty}^{\infty} y^2 dF_Y(y) \\ &= \sum_{i=1}^{\infty} \frac{1}{i^2} \int_{|y| \leq i} y^2 dF_Y(y) \\ &= \sum_{i=1}^{\infty} \frac{1}{i^2} \int_{|x| \leq i} x^2 dF_X(x) && \text{as } Y_i = X_i \text{ if } |Y_i| = |X_i| \leq i \\ &= \sum_{i=1}^{\infty} \frac{1}{i^2} \sum_{j=1}^i \int_{j-1 < |x| \leq j} x^2 dF_X(x) \\ &= \sum_{j=1}^{\infty} \int_{j-1 < |x| \leq j} x^2 dF_X(x) \sum_{i=j}^{\infty} \frac{1}{i^2} && \text{write down the elements and see the pattern} \\ &= \sum_{j=1}^{\infty} \int_{j-1 < |x| \leq j} |x|^2 dF_X(x) \sum_{i=j}^{\infty} \frac{1}{i^2} \\ &\leq \sum_{j=1}^{\infty} \int_{j-1 < |x| \leq j} j \cdot |x| dF_X(x) \sum_{i=j}^{\infty} \frac{1}{i^2} \\ &\leq \sum_{j=1}^{\infty} j \int_{j-1 < |x| \leq j} |x| dF_X(x) \frac{C}{j} && \sum_{i=j}^{\infty} \frac{1}{i^2} \leq \frac{C}{j} \text{ for some } C \\ &= C \sum_{j=1}^{\infty} \int_{j-1 < |x| \leq j} |x| dF_X(x) \\ &\leq C \sum_{j=1}^{\infty} \int_{j-1 \leq |x| \leq j} |x| dF_X(x) \\ &= C \int_{0 \leq |x| < \infty} |x| dF_X(x) \\ &= C \mathbb{E}[|X_1|] < \infty && \text{implies by } \mathbb{E}[X_1] < \infty \end{aligned}$$

Therefore, we have $N^{-1} \sum_{i=1}^N (Y_i - \mathbb{E}[Y_i]) \rightarrow_{as} 0$ (note that $\mu_i = \mathbb{E}[Y_i - \mathbb{E}[Y_i]] = 0$) by Kolmogorov Theorem 1.

Note that $\mathbb{E}[Y_i] = \int_{-\infty}^{\infty} y dF_Y(y) = \int_{|y| \leq i} y dF_Y(y) = \int_{0 \leq |x| \leq i} x dF_X(x)$. Thus, $\lim_{i \rightarrow \infty} \mathbb{E}[Y_i] = \lim_{i \rightarrow \infty} \int_{0 \leq |x| \leq i} x dF_X(x) = \mathbb{E}[X_1]$. Therefore,

$$\begin{aligned} N^{-1} \sum_{i=1}^N (Y_i - \mathbb{E}[Y_i]) &\rightarrow_{as} 0 \\ N^{-1} \sum_{i=1}^N Y_i &\rightarrow_{as} N^{-1} \sum_{i=1}^N \mathbb{E}[Y_i] \rightarrow \frac{N}{N} \mathbb{E}[X_1] = \mathbb{E}[X_1] \end{aligned}$$

as $N \rightarrow \infty$. Now we show that $\{X_i\}_{i=1}^\infty$ and $\{Y_i\}_{i=1}^\infty$ are equivalent,

$$\sum_{i=1}^{\infty} Pr(X_i \neq Y_i) = \sum_{i=1}^{\infty} Pr(|X_i| > i) = \sum_{i=1}^{\infty} Pr(|X_1| > i) \leq \mathbb{E}[|X_1|] < \infty$$

by Lemma 1 and hypothesis $\mathbb{E}[X_1] < \infty$. Thus, $N^{-1} \sum_{i=1}^N X_i \rightarrow_{as} \mathbb{E}[X_1]$

Q.E.D.

3.2 Proof of Weak Law of Large Numbers

3.2.1 Levy's Continuity Theorem

Let $\phi_j(t)$ be the characteristic function of X_j ,

(a) $X_j \rightarrow_d X \implies \phi_j(t) \rightarrow \phi(t)$ pointwise in t where $\phi(t)$ is the characteristic function of X ;

(b) $\phi_j(t) \rightarrow \phi(t)$ pointwise in t and $\phi(0)$ is continuous $\implies \phi(t)$ is a characteristic function of a random variable, say X , and $X_j \rightarrow_d X$.

Proof:

(a) can be proved by a direct application of Portmanteau Lemma (Lemma 2.2 (ii) of Van der Vaart (1998)) as $\exp(itz)$ is a bounded and continuous function of z . See Rao (1973, p.119) for the proof of (b) which applies Helly Lemma. Q.E.D.

Remark: This theorem is also a key ingredient in the proof of Lindeberg-Levy Central Limit Theorem.

3.2.2 Khinchine's Theorem

Let $\{X_i\}_{i=1}^\infty$ be a sequence of i.i.d. random variables, we have

$$\mu := \mathbb{E}[X_1] < \infty \implies N^{-1} \sum_{i=1}^N X_i \rightarrow_p \mu$$

Proof:

(This is from the Proposition 2.16 of Van der Vaart (1998)) Let $\phi(t)$ be the characteristic function of X_i . The first-order Taylor expansion of $\phi(t)$ centered at 0 is

$$\begin{aligned} \phi(t) &= \phi(0) + \phi'(0)t + o(t) \\ &= 1 + i\mu t + o(t) \end{aligned} \quad \mu < \infty \implies \phi'(0) = i\mu$$

Therefore,

$$\begin{aligned} \mathbb{E}_X[\exp(itN^{-1} \sum_{j=1}^N X_j)] &= \mathbb{E}_X[\prod_{j=1}^N \exp(i \frac{t}{N} X_j)] \\ &= \prod_{j=1}^N \mathbb{E}_X[\exp(i \frac{t}{N} X_j)] && \text{by independence of } X_i \text{ and Fubini's Theorem} \\ &= (\phi(t/N))^N && \text{by identical distribution of } X_i \\ &= (1 + i\mu \frac{t}{N} + o(\frac{t}{N}))^N && \text{by Taylor expansion above} \\ &\rightarrow (\exp(i\mu \frac{t}{N}))^N && \text{using } \ln(1+x) \approx x \text{ by Taylor expansion} \\ &= \exp(it\mu) \\ &= \mathbb{E}_\mu[\exp(it\mu)] && \text{as } \mu \text{ is a constant} \end{aligned}$$

Note that $\mathbb{E}_\mu[\exp(it\mu)]$ is continuous at $t = 0$. Thus, $N^{-1} \sum_{j=1}^N X_j \rightarrow_d \mu$ by Levy's Continuity Theorem. As $X \rightarrow_p Y \iff X \rightarrow_d Y$ if Y is a constant, we have $N^{-1} \sum_{j=1}^N X_j \rightarrow_p \mu$ Q.E.D.

Remark 1: This can also be proved without using Levy's Continuity Theorem e.g., see Rao (1973, p.112-113)

Remark 2: With the same hypothesis of Strong Law of Large Numbers i.e., i.i.d. $\{X_j\}_j$ and finite $\mathbb{E}[X_1]$, Weak Law of Large Numbers says that the sample average only converges to $\mathbb{E}[X_1]$ in probability, instead of almost surely. In general, $X \rightarrow_{as} Y \implies X \rightarrow_p Y$. So, the Weak Law says something that is weaker. The proof of it is less involved as a result.

Remark 3: Sample average converging in probability to the first moment of X_j is implied by Chebyshev's inequality. However, it requires a finite variance of X_j (but does not require the independence of $\{X_j\}_j$).

3.2.3 Chebyshev's Inequality

For any random variable X with $\mu := \mathbb{E}[X]$ and $\sigma^2 := \text{Var}(X) < \infty$. For any $\lambda > 0$,

$$\Pr(|X - \mu| \geq \lambda\sigma) \leq \frac{1}{\lambda^2}$$

If we set $\lambda = \varepsilon/\sigma$, we have

$$\Pr(|X - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}$$

Proof:

(This neat proof is from Rao (1973, p.95))

$$\begin{aligned} \sigma^2 &:= \text{Var}(X) = \int_{-\infty}^{\infty} (x - \mu)^2 dF_X(x) \\ &= \int_{-\infty}^{\mu - \lambda\sigma} (x - \mu)^2 dF_X(x) + \int_{\mu - \lambda\sigma}^{\mu + \lambda\sigma} (x - \mu)^2 dF_X(x) + \int_{\mu + \lambda\sigma}^{\infty} (x - \mu)^2 dF_X(x) \\ &\geq \int_{-\infty}^{\mu - \lambda\sigma} (x - \mu)^2 dF_X(x) + \int_{\mu + \lambda\sigma}^{\infty} (x - \mu)^2 dF_X(x) \\ &\geq \int_{-\infty}^{\mu - \lambda\sigma} (\mu - \lambda\sigma - \mu)^2 dF_X(x) + \int_{\mu + \lambda\sigma}^{\infty} (\mu + \lambda\sigma - \mu)^2 dF_X(x) \quad \text{the inequality holds because of the squares} \\ &= \lambda^2 \sigma^2 \int_{-\infty}^{\mu - \lambda\sigma} dF_X(x) + \lambda^2 \sigma^2 \int_{\mu + \lambda\sigma}^{\infty} dF_X(x) \\ &= \lambda^2 \sigma^2 \left(\int_{-\infty}^{\mu - \lambda\sigma} dF_X(x) + \int_{\mu + \lambda\sigma}^{\infty} dF_X(x) \right) \\ &= \lambda^2 \sigma^2 \Pr(|X - \mu| \geq \lambda\sigma) \end{aligned}$$

Q.E.D.

Remark 1: This holds for any random variables drawn from any sampling schemes e.g., i.i.d., i.n.i.d., dependence, etc.

Remark 2: If μ is not $\mathbb{E}[X]$, the inequality still holds if σ^2 is defined as $\mathbb{E}[(X - \mu)^2]$ (not a variance as μ is not a mean).

3.2.4 Chebyshev's Theorem

For any sequence of random variables $\{X_i\}_{i=1}^{\infty}$ with $\mu_i := \mathbb{E}[X_i]$ and $\sigma_i^2 := \text{Var}(X_i) < \infty$ and $\text{Cov}(X_i, X_j) = 0$ for $i \neq j$, we have

$$\frac{\sum_{i=1}^N \sigma_i^2}{N^2} \rightarrow 0 \implies N^{-1} \sum_{i=1}^N X_i \rightarrow_p N^{-1} \sum_{i=1}^N \mu_i$$

as $N \rightarrow \infty$

Proof:

First, note that $Var(N^{-1} \sum_{i=1}^N X_i) = N^{-2} Var(\sum_{i=1}^N X_i) = N^{-2} \sum_{i=1}^N \sigma_i^2$ because the covariances are zero by the hypothesis.

By Chebyshev's Inequality and the hypothesis, we have for any $\varepsilon > 0$,

$$Pr(|N^{-1} \sum_{i=1}^N X_i - N^{-1} \sum_{i=1}^N \mu_i| \geq \varepsilon) \leq \frac{\sum_{i=1}^N \sigma_i^2}{N^2 \varepsilon} \rightarrow 0$$

as $N \rightarrow \infty$

Q.E.D.

Remark 1: This holds for any sequence of random variables drawn from i.i.d. or i.n.i.d. sampling schemes. If the identical distribution is assumed, $N^{-1} \sum_{i=1}^N X_i \rightarrow_p \mu$ as $\sigma^2/N \rightarrow 0$ and $\sigma^2 < \infty$ as $N \rightarrow \infty$. Compared with Khinchine's Theorem, it strengthens the moment condition from the finite first moment to the finite second moment, but it weakens from independence to zero covariances.